Track A: model and verification

Scope: capacity-constrained clustering only. Quantum optimization returns groups; all routing is exact classical evaluation with those groups fixed. No quantum speedup claim.

Binary x_ik assigns each customer exactly once. Minimize the sum of Euclidean customer-pair distances within each vehicle, subject to load <= capacity. Unused vehicles are allowed. The depot is fixed at (50, 50) and is absent from compactness.

The target QUBO divides distances by the largest customer distance, adds assignment squares and binary-slack capacity squares. With U the normalized certificate cost, A = B = U + 1 proves feasibility of every target ground state. The certificate is not a solver warm start. Pair coefficients are twice off-diagonal matrix entries.

Variable counts (assignment / textbook binary slack / implemented): S 8 / 14 / 14; M 30 / 45 / 45; L 60 / 80 / 80. No extra auxiliaries; all pair couplings are nonnegative, but diagonal coefficients generally differ.

Evidence: 10 exact clustering/full-CVRP reference pairs; 10 exhaustive 14-bit QUBO validations (10 passed). Independent S CPLEX cross-checks agree for both references on 10/10 instances.

The separate docs/MODEL.md gives the original CVRP arc/assignment formulation, connectivity constraints, penalty expansion and native Hamiltonian derivation.

Instances, embedding and pulse

10 explicit S demand vectors with (N,K,Q)=(4,2,7). The input in inputs/seeds_s.json contains N, K, Q, demand_range, demands, customer coordinates and depot. Fixed coordinates are read directly from the input. Feasibility witnesses are computed internally, not supplied as input.

References use exhaustive subset dynamic programming, with exact Held-Karp tour costs for every candidate group. Capacity-filtered set partitions select groups and tours jointly for full CVRP, independently of compactness. Vehicle labels are quotiented. S additionally counts tour optima modulo reversal.

Embedding uses a legal triangular grid, seeded greedy atom-label swaps and a least-squares positive interaction scale. Every pair is retained, including unwanted interactions where the target is zero. Device validation enforces 5 um spacing, 38 um radius, 80 atoms and all installed AnalogDevice controls.

Native energy at zero drive is -delta*sum(z) + sum(C6/r^6*z_a*z_b). C6 = 865723.02 rad/us*um^6 from Pulser. The fitted scale and scaled constant are saved. Global detuning fits the mean diagonal only: this is not an exact QUBO.

Pair error is ||J_real - s J_target||_F / ||s J_target||_F over all off-diagonal pairs. Linear error is ||h_real - s h||_2 / ||s h||_2. Embedding fidelity is characterized by these errors, not by a claimed exact mapping.

Calibration: {'duration_ns': 1000, 'amplitude': 8.0, 'initial_detuning': -8.0}. The two fixed candidate sweeps use 1000/2000 ns, peak amplitude 8 rad/us, initial detuning -8 rad/us and zero final amplitude. Settings freeze after S-0. Calibration uses a benchmark instance, so it is not held-out validation.

Global-only primary execution, noiseless emu-sv on CPU with dt=10 ns, zero interaction cutoff and recorded shot seeds. Only S is executed. M/L, MPS scaling, DMM and QPU operation are deferred.

Observed results and comparison

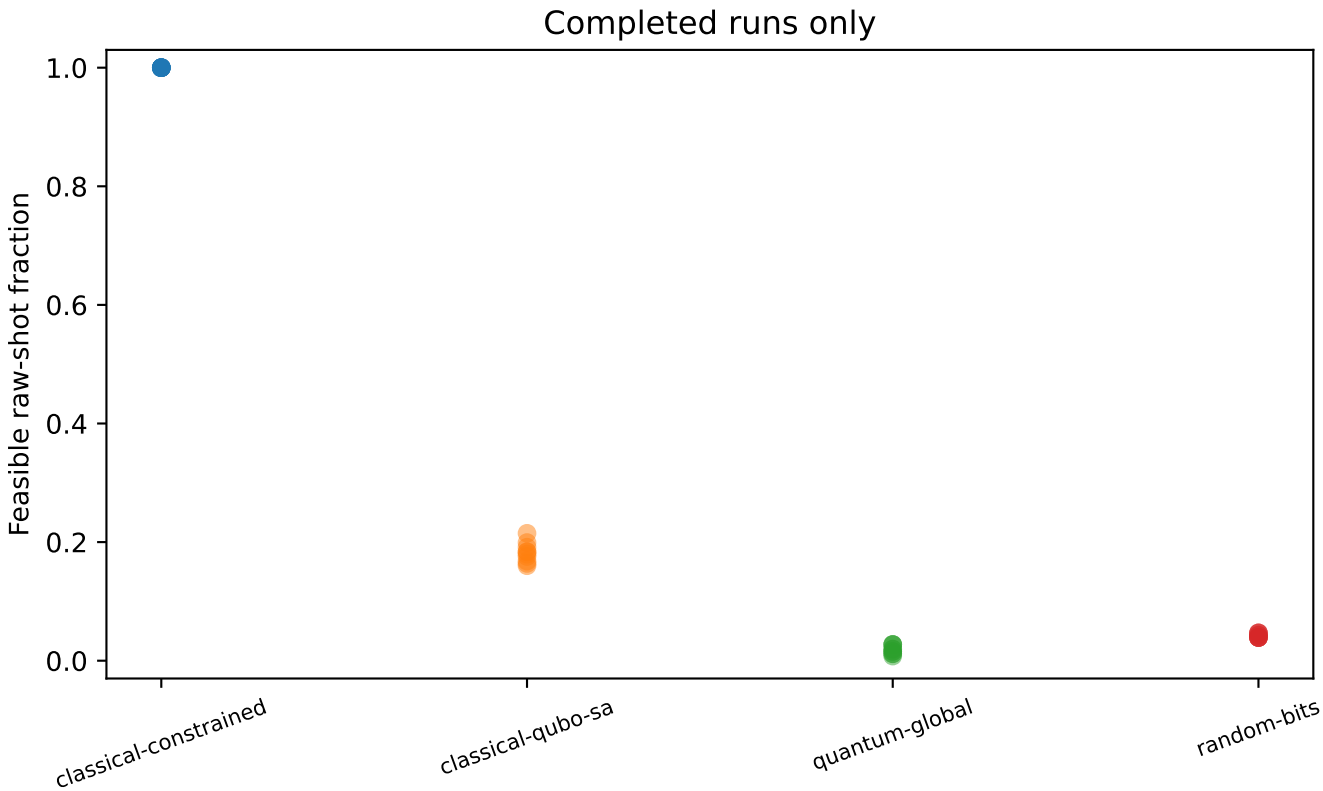
Quantum run statuses: {'completed': 10}. All statuses, failure messages, actual shots, reference values and gaps are preserved in results.csv.

Feasibility is measured against the original assignment and capacity constraints, not against slack-bit residuals. Reference-hit probabilities use all raw shots as denominator. Gap thresholds: 0.1%, 1%, 5%. Undefined gaps remain missing; negative best-known gaps are never clipped.

The best compactness gap and best CVRP gap may come from different sampled partitions. Each receives exact fixed-group depot tours; routing never changes groups or feeds back into the quantum optimizer.

Classical constrained results are deterministic exact solutions repeated to the common shot budget, not independent stochastic samples. Classical QUBO comparisons use independent 400-step Metropolis chains; random baselines use uniform raw bits. Both use identical decoding and feasibility filtering.

Reference, clustering, embedding, emulation and routing runtimes are separated. Reference DP is offline. The deterministic clustering baseline independently enumerates feasible partitions; this comparison measures quality rather than speedup.



What did not work, limits and AI use

The dense slack QUBO cannot generally be reproduced by 2D repulsive interactions with one shared detuning. Positive coupling signs alone do not establish embeddability. Large saved errors are physical-model approximation failures, not failures of the proven target penalty or exact references.

Completed quantum S runs: 10; runs with no feasible raw shots: 0. M/L are intentionally out of scope. Resource-limited or failed runs have no reported success metrics.

Resource limits bound each isolated state-vector emulator by wall time and address space. Time-step convergence and noise robustness are not established by this baseline. No scaling or hardware performance claims are made.

Only one geometry is covered. The supplied vectors share a demand multiset, although the simplified input also accepts different feasible vectors. Low feasible rates on uniform bitstrings are expected for one-hot assignment. A small pulse calibration and local swap layout are baselines, not an exhaustive embedding/control optimization. There is no classical repair of quantum shots.

Reproduction: uv run --locked python -m track_a.common.run. Saved evidence is reassembled with --stage results without rerunning clustering or quantum jobs. The three notebooks show executable core algorithms with beginner-friendly explanations and saved-result views.

AI-use disclosure: an AI assistant using Copilot SDK in VS Code implemented the shared modules, notebooks, derivations and automated report. The submitted numbers come from actual local solvers/emulators and saved raw outputs, not generated claims. Human review remains necessary.